CAPÍTULO 1
1.4 Exercícios — pág 18
(1 a 11)

Observação para o leitor: os gráficos apresentados foram construídos em softwares livres. Em geral os de duas dimensões foram realizados com o Graph (http://www.padowan.dk) e os gráficos de três dimensões com o Winplot (http://math.exeter.edu/rparris).

1. Encontrar uma função de várias variáveis que nos dê:

a) O comprimento de uma escada apoiada como na figura 1.35.

$$H^2 + L^2 = C^2$$
$$C = \sqrt{H^2 + L^2}$$
$$C(H, L) = \sqrt{H^2 + L^2}\ .$$

b) O volume de água necessário para encher uma piscina redonda de *x* metros de raio e *y* metros de altura.

$$V(x, y) = \pi x^2 y\ .$$

c) A quantidade de rodapé, em metros, necessária para se colocar numa sala retangular de largura *a* e comprimento *b*.

$$f(a, b) = 2a + 2b\ .$$

d) A quantidade, em metros quadrados, de papel de parede necessária para revestir as paredes laterais de um quarto retangular de *x* metros de largura, *y* metros de comprimento, se a altura do quarto é *z* metros.

$$f(x, y, z) = 2yz + 2xz\ .$$

e) O volume de um paralelepípedo retângulo de dimensão *x*, *y* e *z*.

$$V(x, y, z) = x\,y\,z\ .$$

f) A distância entre dois pontos $P(x, y, z)\ e\ Q(u, v, w)$.

$$d((x, y, z), (u, v, w)) = \sqrt{(x-u)^2 + (y-v)^2 + (z-w)^2}\ .$$

g) A temperatura nos pontos de uma esfera se ela, em qualquer ponto, é numericamente igual à distância do ponto ao centro da esfera.

Para uma esfera centrada em (x_0, y_0, z_0) temos:

$T(x,y,z) = \sqrt{(x-x_0)^2 + (y-y_0)^2 + (z-z_0)^2}$.

2. Uma loja vende certo produto P de duas marcas distintas A e B. A demanda do produto com marca A depende do seu preço e do preço da marca competitiva B. A demanda do produto com marca A é
$D_A = 1300 - 50x + 20y$ unidades/mês
e do produto com marca B é
$D_B = 1700 + 12x - 20y$ unidades/mês
onde *x* é o preço do produto A e *y* é o preço do produto B.

Escrever uma função que expresse a receita total mensal da loja, obtido com a venda do produto P.

Receita = (número de unidades A por mês) x + (número de unidades B por mês) y

$$\begin{aligned} R(x,y) &= (1300 - 50x + 20y)x + (1700 + 12x - 20y)y \\ &= 1300x - 50x^2 + 20xy + 1700y + 12xy - 20y^2 \\ &= 1300x + 1700y - 50x^2 - 20y^2 + 32xy. \end{aligned}$$

3. Determinar o domínio e o conjunto imagem das seguintes funções:

a) $z = 3 - x - y$

$D(z) = \Re^2$
$\mathrm{Im}(z) = \Re$

b) $f(x,y) = 1 + x^2 + y^2$

$D(f) = \Re^2$
$\mathrm{Im}(f) = [1, \infty)$

c) $z = \sqrt{9 - (x^2 + y^2)}$
Temos que:
$9 - (x^2 + y^2) \geq 0$
$x^2 + y^2 \leq 9$
Assim,

$D(z) = \{(x, y) \in \Re^2 / x^2 + y^2 \leq 9\}$
$\text{Im}(z) = [0,3]$

d) $w = e^{x^2+y^2+z^2}$

$D(w) = \Re^3$
$\text{Im}(w) = [1,+\infty)$

e) $f(x, y, z) = \sqrt{x^2 + y^2 + z^2}$
Temos que:
$x^2 + y^2 + z^2 \geq 0$
Assim,
$D(f) = \{(x, y, z) \in \Re^3 / x^2 + y^2 + z^2 \geq 0\} = \Re^3$
$\text{Im}(f) = [0,\infty) = \Re_+$

f) $f(x, y) = 2x + 5y - 4$
$D(f) = \Re^2$
$\text{Im}(f) = \Re$

g) $z = x^2 + y^2 - 2$
$D(z) = \Re^2$
$\text{Im}(z) = [-2,+\infty)$

h) $f(x, y) = 2x^2 + 5y$
$D(f) = \Re^2$
$\text{Im}(f) = \Re$

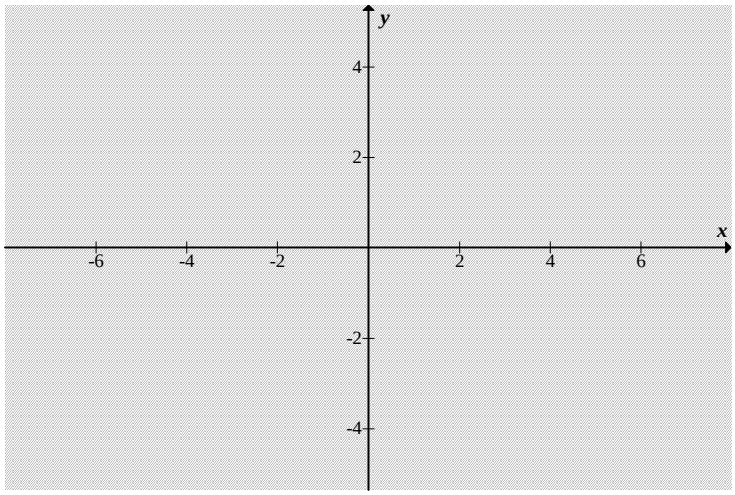
i) $w = 4 + x^2 + y^2$
$D(w) = \Re^2$
$\text{Im}(w) = [4,+\infty)$

j) $f(x, y) = 4 - x^2 - y^2$
$D(f) = \Re^2$
$\text{Im}(f) = (-\infty,4]$

4. Determinar o domínio das seguintes funções e representar graficamente:

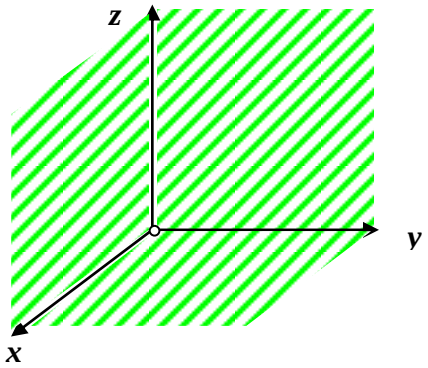
a) $z = xy$

$D(z) = \Re^2$



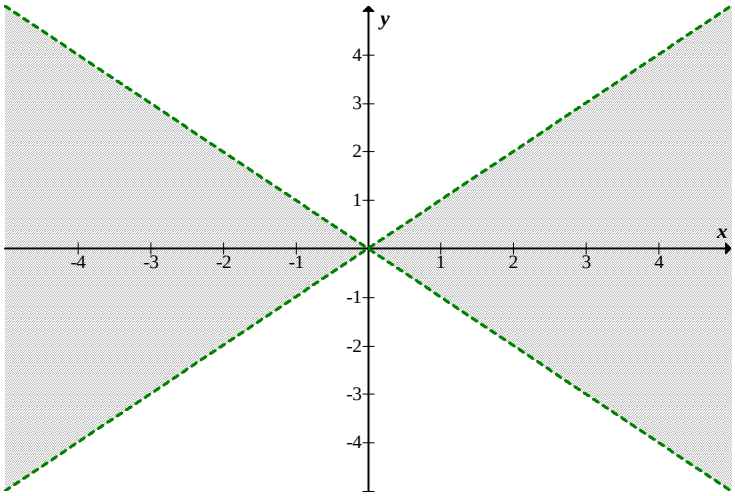
b) $w = \dfrac{1}{x^2 + y^2 + z^2}$

$D(w) = \{(x, y, z) \in \Re^3 / (x, y, z) \neq (0,0,0)\}$



c) $z = \dfrac{1}{\sqrt{x^2 - y^2}}$

$D(z) = \{(x, y) \in \Re^2 / x^2 - y^2 > 0\} = \{(x, y) \in \Re^2 / |x| > |y|\}$

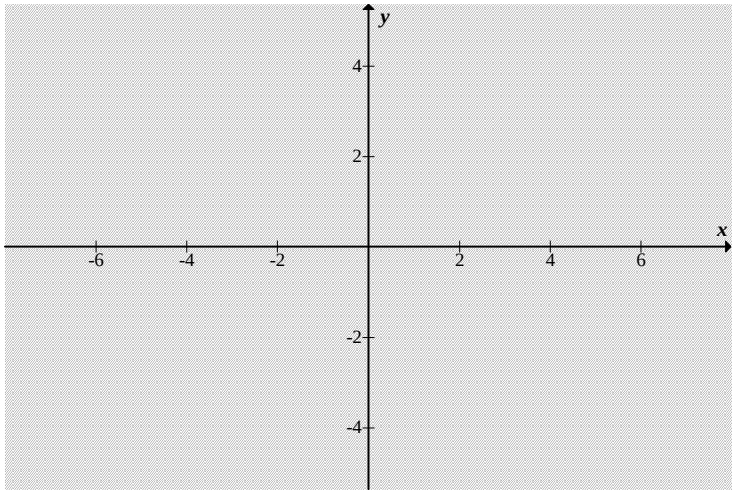


d) $z = \dfrac{x}{y^2 + 1}$

$$D(z) = \{(x, y) \in \Re^2 \,/\, y^2 + 1 \neq 0\}$$

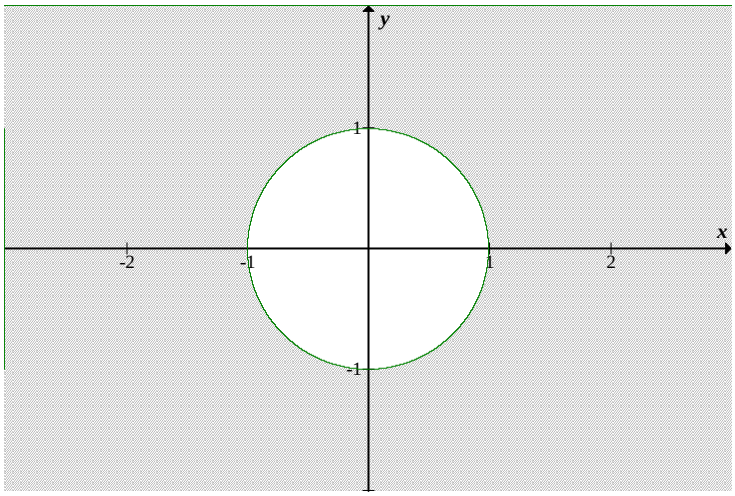
Como $y^2 + 1$ é sempre $\neq 0$, temos que:

$$D(z) = \Re^2$$



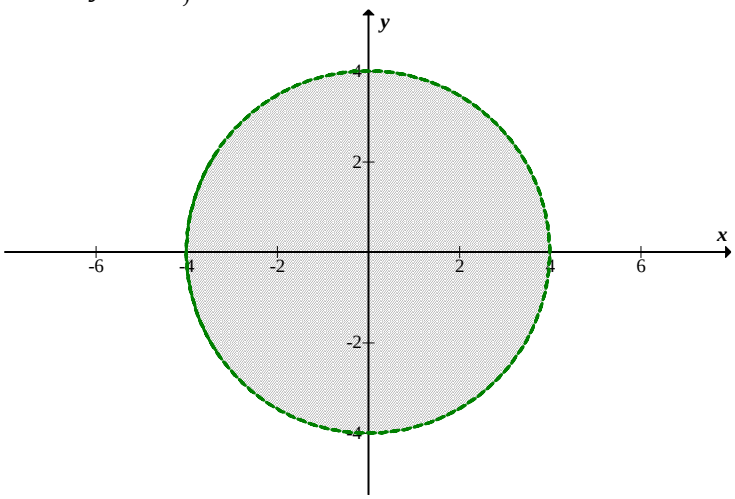
e) $z = \sqrt{x^2 + y^2 - 1}$

$$\begin{aligned} D(z) &= \{(x, y) \in \Re^2 \,/\, x^2 + y^2 - 1 \geq 0\} \\ &= \{(x, y) \in \Re^2 \,/\, x^2 + y^2 \geq 1\} \end{aligned}$$



f) $z = \ln\left(4 - \sqrt{x^2 + y^2}\right)$

$$D(z) = \left\{(x, y) \in \Re^2 / 4 - \sqrt{x^2 + y^2} > 0\right\}$$
$$= \left\{(x, y) \in \Re^2 / x^2 + y^2 < 16\right\}$$

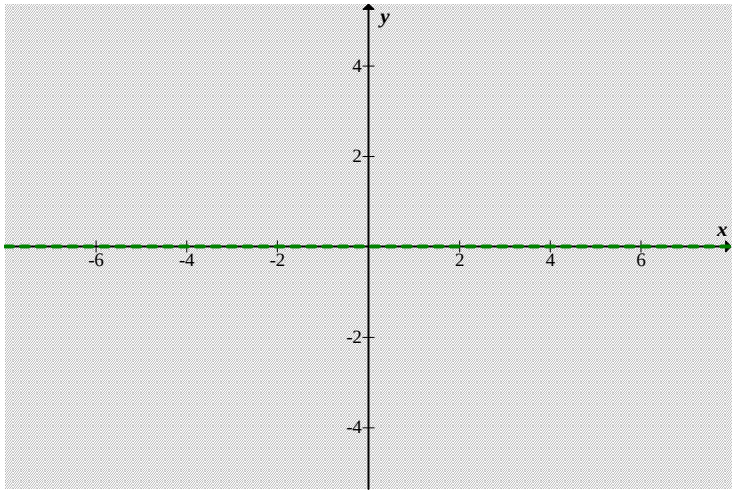


g) $z = \ln\left(\dfrac{\sqrt{x^2 + y^2} - x}{\sqrt{x^2 + y^2} + x}\right)$

$$D(z) = \left\{(x, y) \in \Re^2 / (x, y) \neq (x, 0) \; com \; x \in \Re\right\}$$

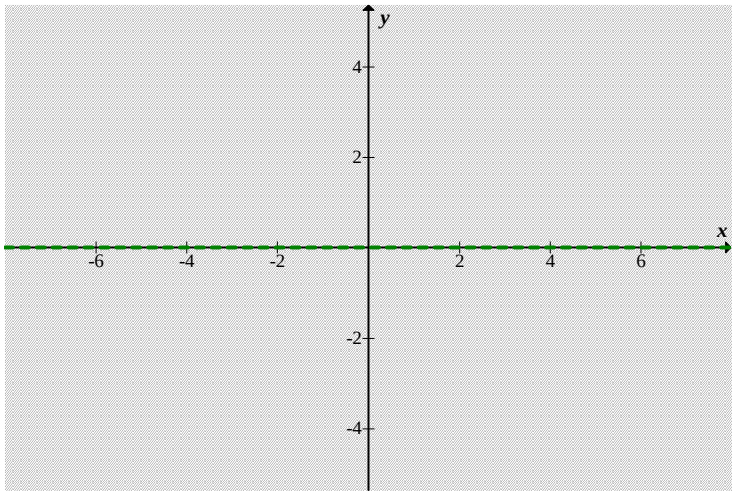
ou

$\Re^2$ - conjunto dos pontos do eixo dos *x*.



h) $z = e^{\frac{x}{y}}$

$$D(z) = \{(x, y) \in \Re^2 \,/\, y \neq 0\}$$

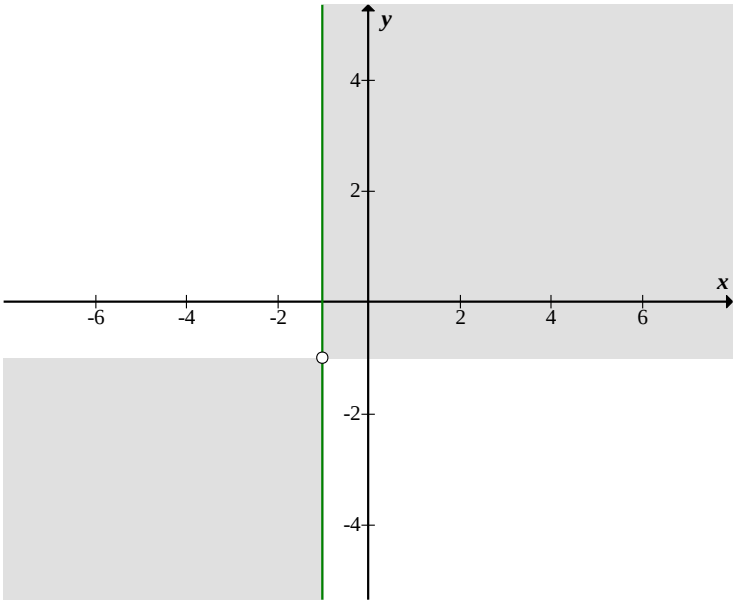


i) $y = \sqrt{\dfrac{1+x}{1+z}}$

$$D(y) = \left\{(x, y) \in \Re^2 \,/\, \frac{1+x}{1+z} \geq 0 \;\; e \;\; 1+z \neq 0\right\}$$

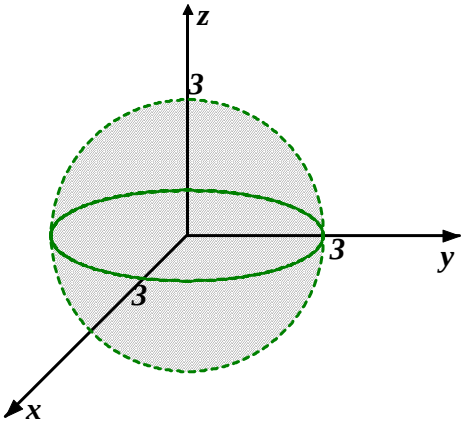
1° caso: $1+x \geq 0 \;\; e \;\; 1+z > 0$
2° caso: $1+x \leq 0 \;\; e \;\; 1+z < 0$

Logo: 1) $x \geq -1 \;\; e \;\; z > -1$ ou
2) $x \leq -1 \;\; e \;\; z < -1$



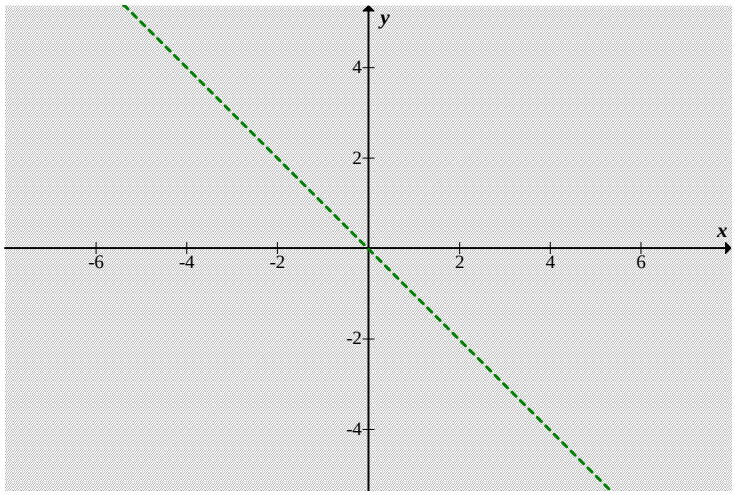
j) $w = \dfrac{1}{\sqrt{9 - x^2 - y^2 - z^2}}$

$$D(w) = \{(x, y, z) \in \Re^3 / 9 - x^2 - y^2 - z^2 > 0\}$$
$$= \{(x, y, z) \in \Re^3 / x^2 + y^2 + z^2 < 9\}$$



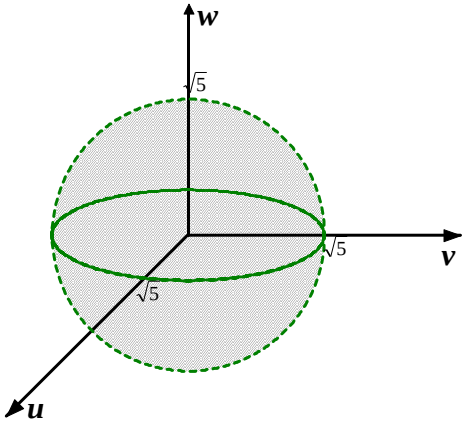
k) $z = \dfrac{4}{x + y}$

$$D(z) = \{(x, y) \in \Re^2 / y \neq -x\}$$



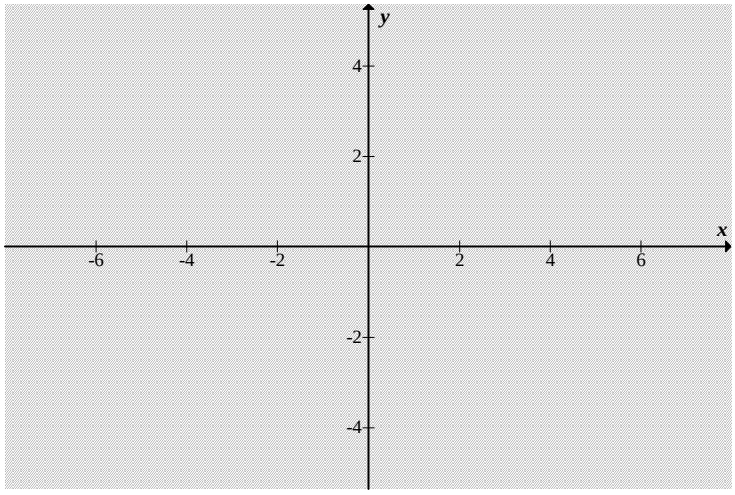
l) $z = \sqrt{5 - u^2 - v^2 - w^2}$

$$D(z) = \{(u, v, w) \in \Re^3 / 5 - u^2 - v^2 - w^2 \geq 0\}$$
$$= \{(u, v, w) \in \Re^3 / u^2 + v^2 + w^2 \leq 5\}$$



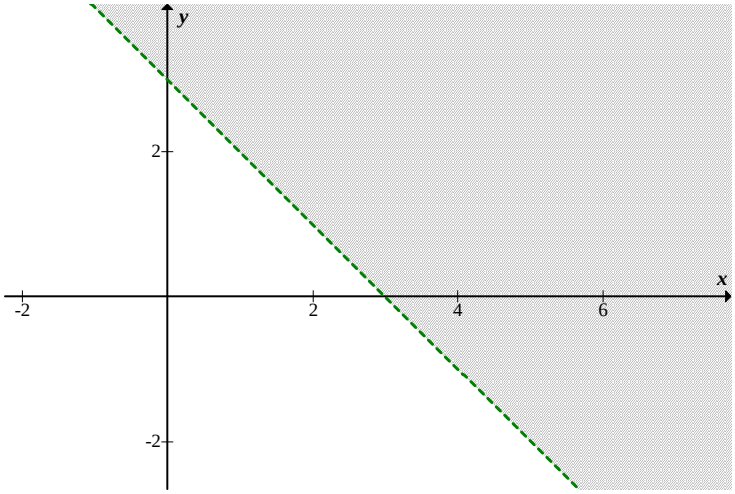
m) $f(x, y) = \sqrt[3]{x^2 + y^2}$

$D(f) = \Re^2$



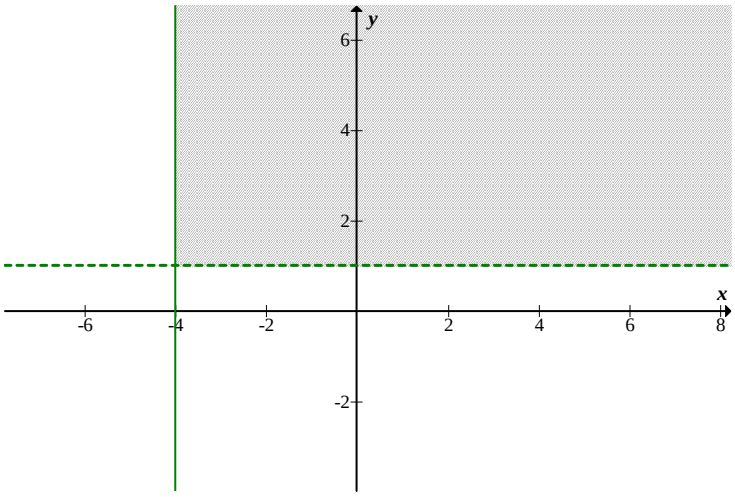
n) $z = \ln(x + y - 3)$

$$D(z) = \{(x, y) \in \Re^2 / x + y - 3 > 0\}$$
$$= \{(x, y) \in \Re^2 / x + y > 3\}$$



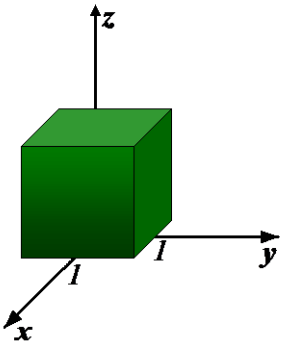
o) $z = \dfrac{\sqrt{x+4}}{\sqrt{y-1}}$

$$D(z) = \{(x, y) \in \Re^2 / x + 4 \geq 0 \;\; e \;\; y - 1 > 0\}$$
$$= \{(x, y) \in \Re^2 / x \geq -4 \;\; e \;\; y > 1\}$$



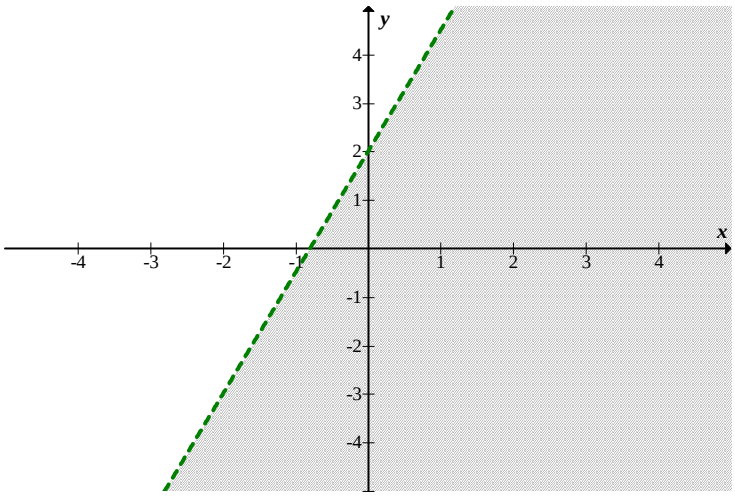
p) $f(x,y,z)=\sqrt{1-x^2}+\sqrt{1-y^2}-\sqrt{1-z^2}$

$$\begin{aligned}
D(f) &= \{(x,y,z)\in\Re^3 / 1-x^2\geq 0\ ,\ 1-y^2\geq 0\ \ e\ \ 1-z^2\geq 0\} \\
&= \{(x,y,z)\in\Re^3 / (1-x)(1+x)\geq 0\ ,\ (1-y)(1+y)\geq 0\ \ e\ \ (1-z)(1+z)\geq 0\} \\
&= \{(x,y,z)\in\Re^3 / (x-1)(x+1)\leq 0\ ,\ (y-1)(y+1)\leq 0\ \ e\ \ (z-1)(z+1)\leq 0\} \\
&= \{(x,y,z)\in\Re^3 / -1\leq x\leq 1\ ,\ -1\leq y\leq 1\ \ e\ \ -1\leq z\leq 1\}
\end{aligned}$$



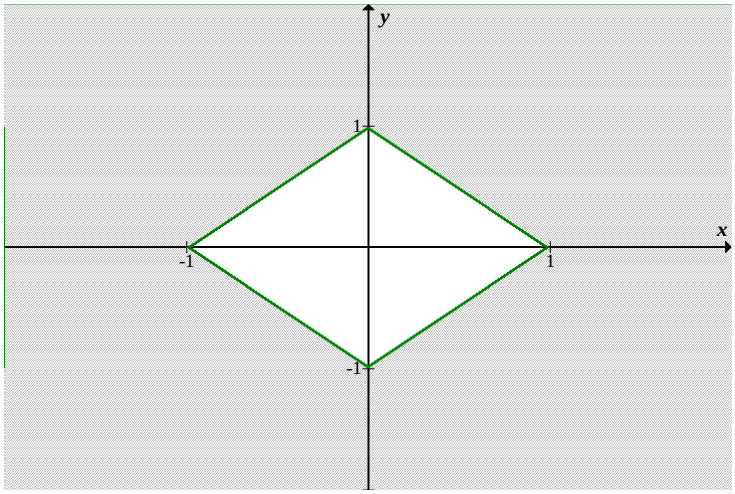
q) $z=\ln(5x-2y+4)$

$D(z)=\{(x,y)\in\Re^2 \mid 5x-2y+4>0\}$



r) $z = \sqrt{|x| + |y| - 1}$

$$D(z) = \{(x, y) \in \Re^2 \mid |x| + |y| - 1 \geq 0\}$$
$$= \{(x, y) \in \Re^2 \mid |x| + |y| \geq 1\}$$



5. A partir da equação dada, definir 2 funções de duas variáveis, determinando seu domínio.

Obs.: podemos obter outras respostas.

a) $y^2 = x^2(9 - x^2) + z$

$$z = y^2 - x^2(9 - x^2)$$
$$y = \sqrt{x^2(9 - x^2) + z}$$

$$D(z) = \Re^2$$
$$D(y) = \{(x,z) \in \Re^2 \mid z \geq x^2(x^2 - 9)\}$$

b) $x^2 + (y-3)^2 + z^2 = 9$

$$z_1 = \sqrt{9 - x^2 - (y-3)^2}$$
$$z_2 = -\sqrt{9 - x^2 - (y-3)^2}$$

$$D(z_1) = D(z_2) = \{(x,y) \in \Re^2 \mid 9 - x^2 - (y-3)^2 \geq 0\}$$
$$= \{(x,y) \in \Re^2 \mid x^2 + (y-3)^2 \leq 9\}$$

c) $l^2 = m^2 + n^2$

$$l_1 = \sqrt{m^2 + n^2}$$
$$l_2 = -\sqrt{m^2 + n^2}$$

$$D(l_1) = D(l_2) = \Re^2$$

6. Dada a função $f(x,y) = \dfrac{x+y}{2x+y}$:

a) Dar o domínio:

$D(z) = \{(x,y) \in \Re^2 \mid 2x + y \neq 0\}$:

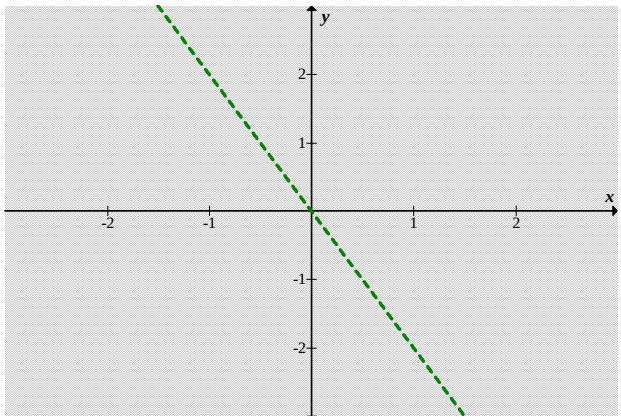
b) Calcular $f(x + \Delta x, y)$

$$f(x+\Delta x, y) = \frac{x + \Delta x + y}{2(x+\Delta x) + y} = \frac{x + \Delta x + y}{2x + 2\Delta x + y}$$

c) Calcular $f(-1,0)$

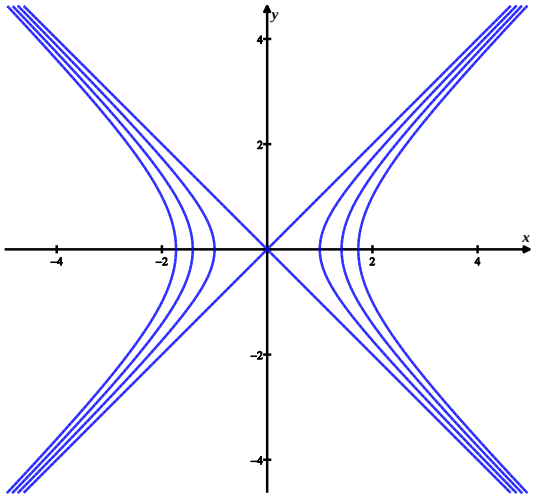
$$f(-1,0) = \frac{-1+0}{2(-1)+0} = \frac{-1}{-2} = \frac{1}{2} \ :$$

d) Fazer um esboço gráfico do domínio:

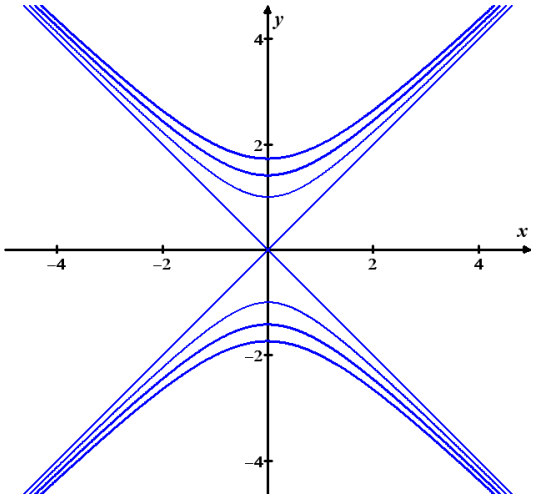


7. Desenhar as curvas de nível C_k para os valores de k dados.

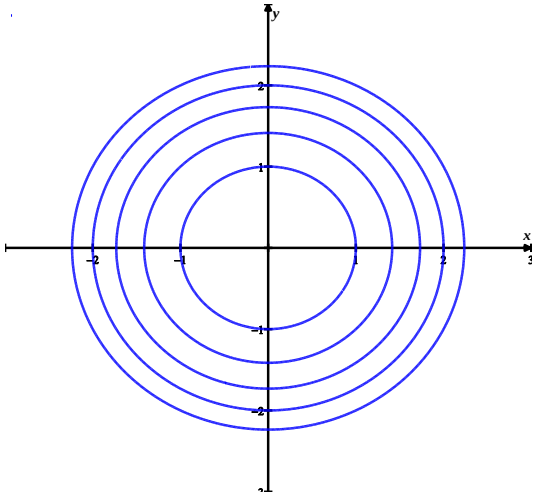
a) $z = x^2 - y^2 \quad ; \quad k = 0,1,2,3$



b) $z = y^2 - x^2 \;\; ; \;\; k = 0,1,2,3$

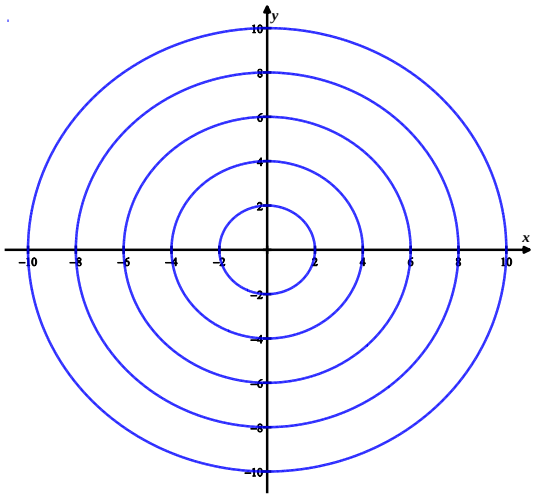


c) $z = 2 - \left(x^2 + y^2\right)$; $k = -3, -2, -1, 0, 1, 2$



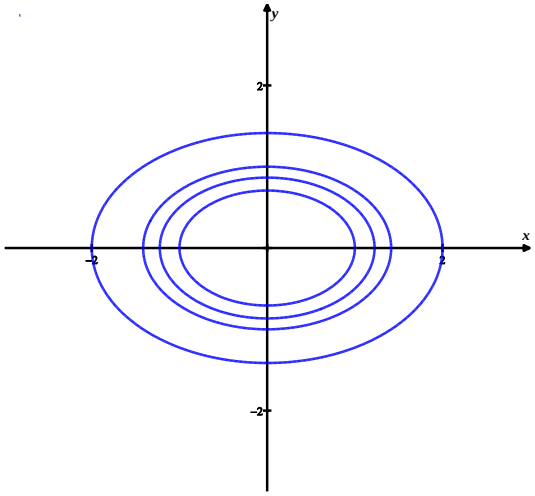
Observamos que para *k=2*, temos uma curva degenerada (*x=y=0)*.

d) $l = \frac{1}{2}\sqrt{m^2 + n^2}$; $k = 0, 1, 2, 3, 4, 5$

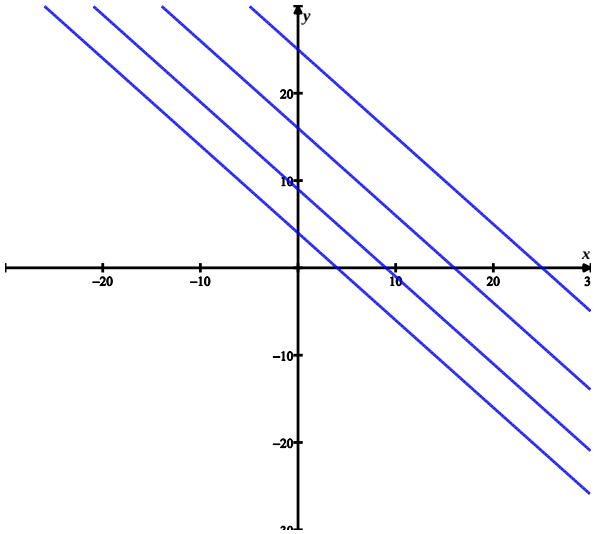


Observamos que para *k=0* temos uma curva de nível degenerada (*m=n=0*).

e) $f(x,y) = 2x^2 + 4y^2$; $k = 2,3,4,8$



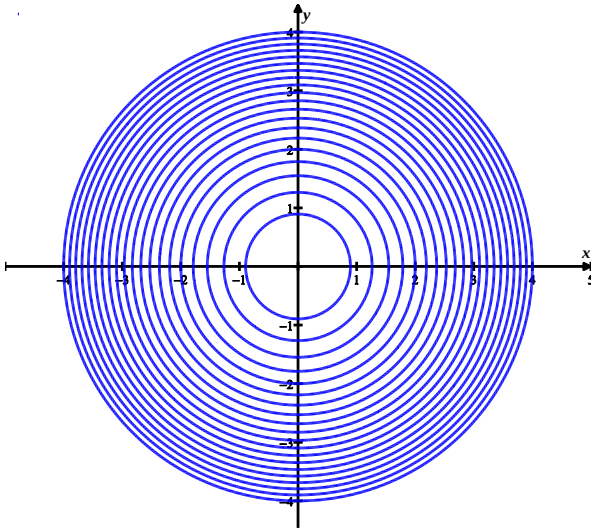
f) $f(x,y) = \sqrt{x+y}$; $k = 5,4,3,2$



Nos exercícios 8 a 10, o conjunto S representa uma chapa plana, e $T(x,y)$, a temperatura nos pontos da chapa. Determinar as isotermas, representando-as geometricamente

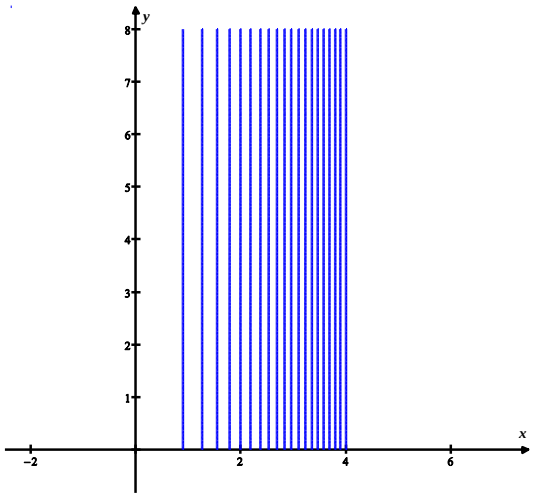
8- $S = \{(x,y) \mid x^2 + y^2 \leq 16\}$; $T(x,y) = x^2 + y^2$

Resposta: circunferências concêntricas $x^2 + y^2 = k \;\; com \;\; 0 \leq k \leq 16$



9 - $S = \{(x,y) / 0 \leq x \leq 4 \text{ , } 0 \leq y \leq 8\}$; $T(x,y) = 4 - x^2$

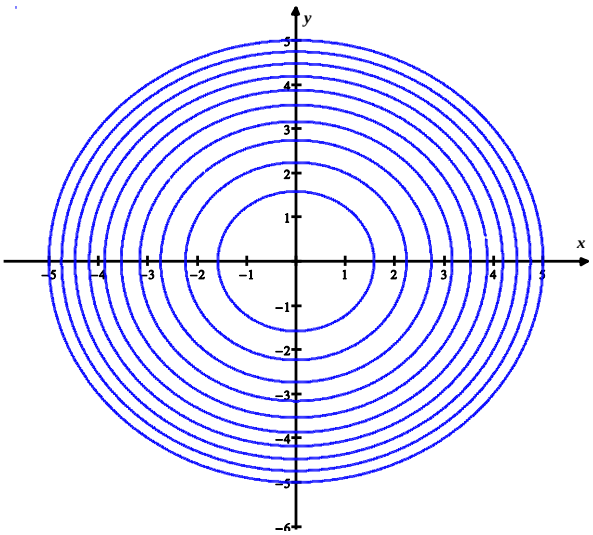
Resposta: segmentos de retas verticais $x = \sqrt{4-k}$, $-12 \leq k \leq 4$



10- $S = \{(x, y) / x^2 + y^2 \leq 25\}$; $T(x, y) = 2(4 - x^2 - y^2)$

$$2(4 - x^2 - y^2) = k$$
$$4 - x^2 - y^2 = \frac{k}{2}$$
$$x^2 + y^2 = 4 - \frac{k}{2}$$

$$0 \leq 4 - \frac{k}{2} \leq 25$$
$$0 \leq 8 - k \leq 50$$
$$-8 \leq -k \leq 42$$
$$8 \geq k \geq -42$$
$$-42 \leq k \leq 8$$

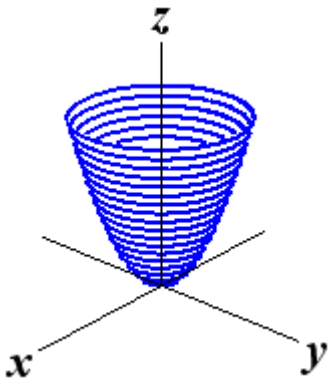


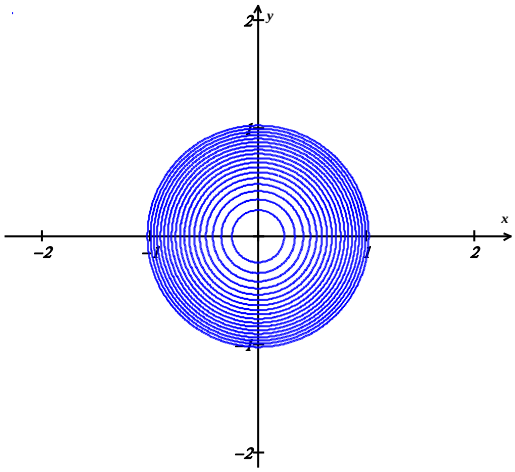
Resposta: circunferências concêntricas $x^2 + y^2 = 4 - \frac{k}{2}$, $-42 \le k \le 8$; para *k=8*, temos uma curva de nível degenerada (*x=y=0*).

11. Desenhar algumas curvas de nível e esboçar o gráfico dos seguintes parabolóides.

a) $z = 2x^2 + 2y^2$

Gráfico da função e das curvas de níveis.





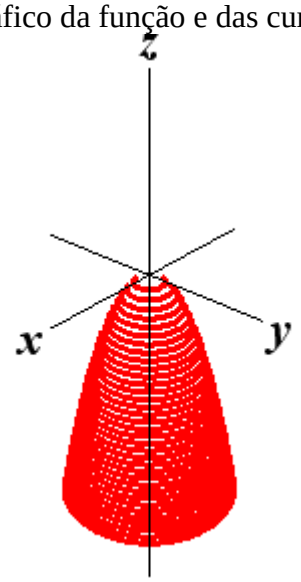
Exemplos de curvas de níveis

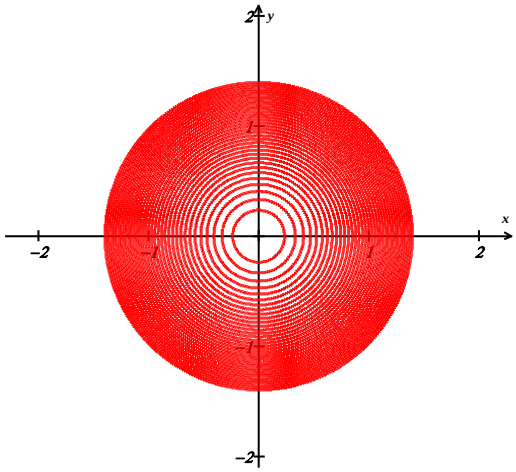
$2 = 2x^2 + 2y^2$ e $1 = 2x^2 + 2y^2$

$x^2 + y^2 = 1$ $\quad$ $x^2 + y^2 = \frac{1}{2}$

b) $z = -2x^2 - 2y^2$

Gráfico da função e das curvas de níveis.





Exemplos de curvas de níveis

$-2 = -2x^2 - 2y^2$

$x^2 + y^2 = 1$

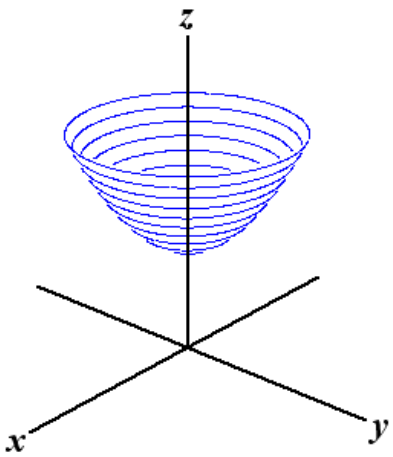
e

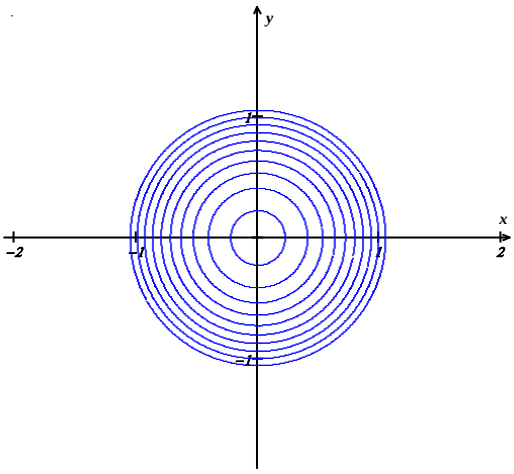
$-1 = -2x^2 - 2y^2$

$x^2 + y^2 = \frac{1}{2}$

c) $z = x^2 + y^2 + 1$

Gráfico da função e das curvas de níveis.





Exemplos de curvas de níveis

$x^2 + y^2 + 1 = 2$

$x^2 + y^2 = 1$

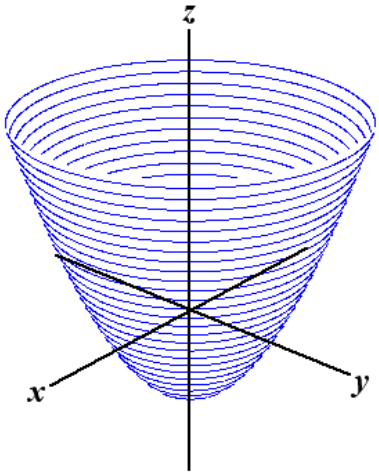
e

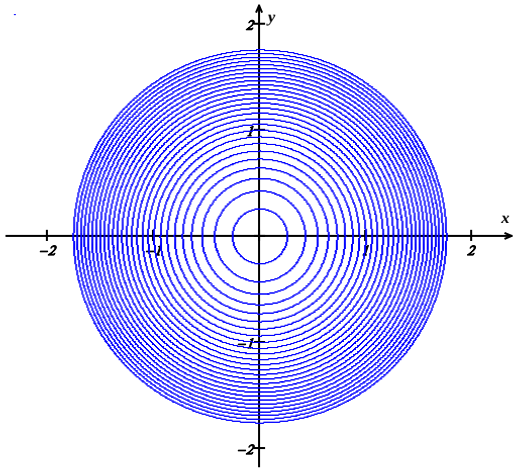
$x^2 + y^2 + 1 = \frac{3}{2}$

$x^2 + y^2 = \frac{1}{2}$

d) $z = x^2 + y^2 - 1$

Gráfico da função e das curvas de níveis.





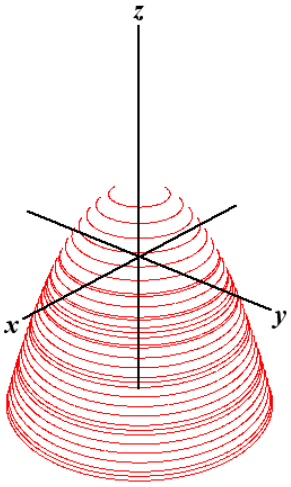
Exemplos de curvas de níveis

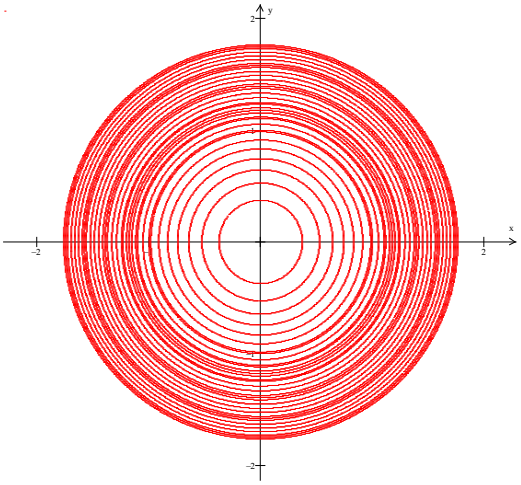
$x^2 + y^2 - 1 = 2$, $x^2 + y^2 - 1 = 1$ e $x^2 + y^2 - 1 = 0$

$x^2 + y^2 = 3$, $x^2 + y^2 = 2$ e $x^2 + y^2 = 1$

e) $z = 1 - x^2 - y^2$

Gráfico da função e das curvas de níveis.





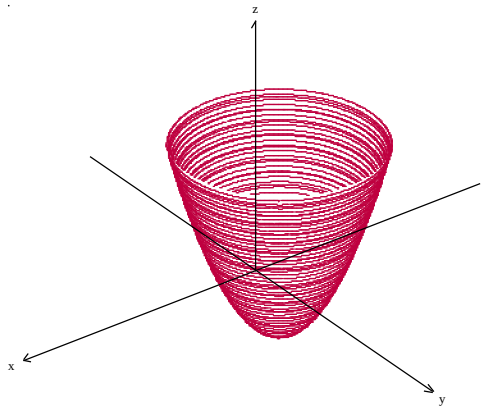
Exemplos de curvas de níveis

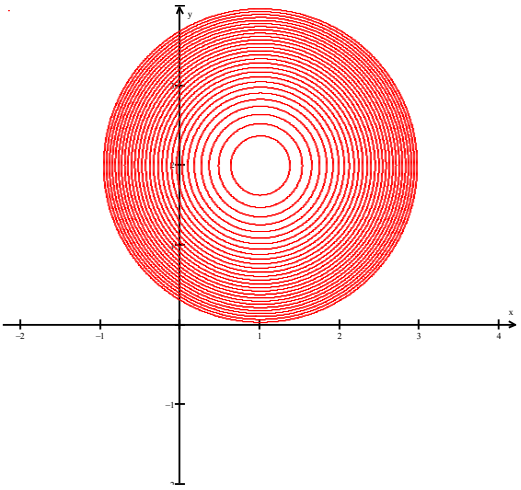
$1 - x^2 - y^2 = 0$, $1 - x^2 - y^2 = -1$ e $1 - x^2 - y^2 = -2$

$x^2 + y^2 = 1$, $x^2 + y^2 = 2$ e $x^2 + y^2 = 3$

f) $z = (x-1)^2 + (y-2)^2$

Gráfico da função e das curvas de níveis.





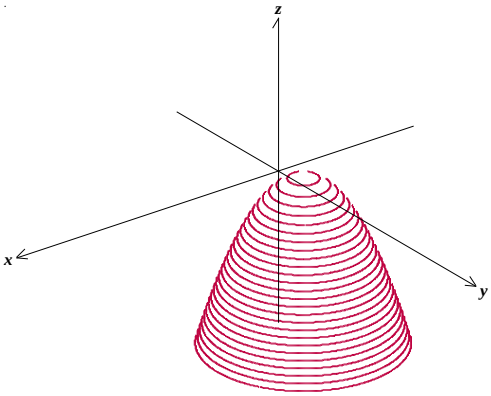
Exemplos de curvas de níveis

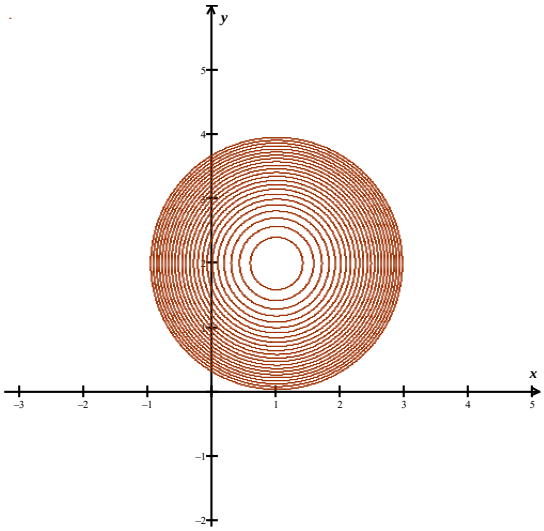
$(x-1)^2 + (y-2)^2 = 2$

$(x-1)^2 + (y-2)^2 = 1$

g) $z = 1 - (x-1)^2 - (y-2)^2$

Gráfico da função e das curvas de nível.





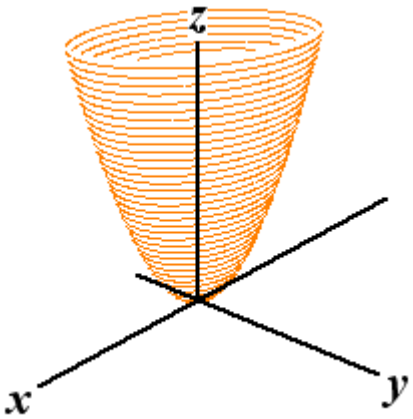
Exemplos de curvas de níveis

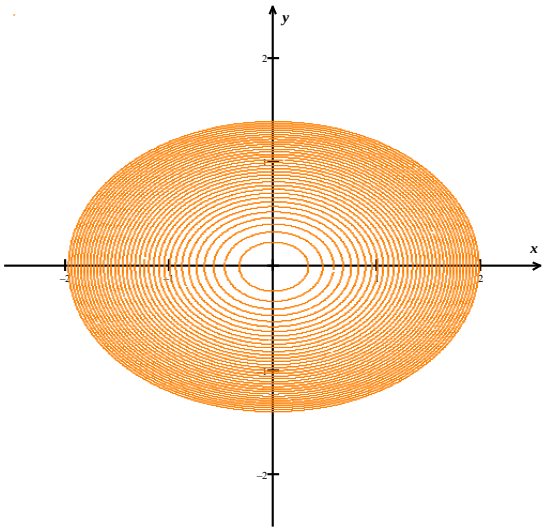
$$-2 = 1 - (x-1)^2 - (y-2)^2$$
$$(x-1)^2 + (y-2)^2 = 3$$

$$0 = 1 - (x-1)^2 - (y-2)^2$$
$$(x-1)^2 + (y-2)^2 = 1$$

h) $z = x^2 + 2y^2$

Gráfico da função e das curvas de níveis.





Exemplos de curvas de níveis

$$1 = x^2 + 2y^2$$
$$\frac{x^2}{1} + \frac{y^2}{\frac{1}{2}} = 1$$

e

$$4 = x^2 + 2y^2$$
$$\frac{x^2}{4} + \frac{y^2}{2} = 1$$